AEA Papers and Proceedings 2023, 113: 310–315
https://doi.org/10.1257/pandp.20231104

Wildfire Insurance, Information, and Self-Protection†

By Judson Boomhower, Meredith Fowlie, and Andrew J. Plantinga*

Wildfire activity has reached crisis proportions across the American West. Contributing factors include climate change, decades of wildfire suppression, and expanding development in the wildland-urban interface. Both the frequency and the intensity of wildfires are projected to increase in the years to come, underscoring the importance of accelerating adaptation efforts.

One form of adaptation is to move populations out of harm's way. To date, development pressures and a shortage of affordable housing have moved people in the opposite direction. In the United States, high-fire-risk areas have been experiencing some of the fastest population growth (Radeloff et al. 2018). Efforts are now under way to slow this trend. But adaptation strategies will be needed for the many millions of households already living in high-risk locations.

Fortunately, there are investments that homeowners can make—such as cutting back trees and vegetation, screening vents, or structural modifications—that reduce the risk of damage in the event of a wildfire (Cohen 2000; Calkin et al. 2014). To date, households appear to be underinvesting in these protective measures (see, e.g., Goolsby et al. 2022 and related surveys). The literature offers several explanations for this underinvestment, including information failures, misperceptions of climate risk, moral hazard caused by insurance coverage and/or government disaster relief, and positive spillovers to neighboring properties that create a divergence between private and social returns on self-protection investments (Kunreuther and Michel-Kerjan 2009; Shafran 2008; Lueck and Yoder 2016; Baylis and Boomhower 2021, 2023).

As wildfire risk and associated damages increase, policymakers are looking for ways to promote cost-effective investments in risk mitigation (termed "self-protection" in the literature). Insurance pricing offers one possible lever. In principle, wildfire insurance prices and eligibility rules could be designed to provide targeted incentives for cost-effective investments. There is mounting policy interest in understanding how this could work in practice.

This paper investigates the interactions between wildfire insurance pricing, information, and private investments in wildfire risk mitigation. Our point of departure is the canonical work of Ehrlich and Becker (1972), who challenged the notion that moral hazard is an inevitable consequence of private insurance provision. Their challenge was predicated on two key assumptions. First, insurers understand the causal relationship between investments in self-protection and damage reduction, and they discount insurance prices commensurately. Second, they assume that households are fully informed about their risk exposure and the extent to which these risks can be mitigated by investments in self-protection.

These two theoretical assumptions find limited empirical support in markets for wildfire insurance. Identifying locations where risk mitigation measures are cost-effective is a complex and data-intensive exercise. Until recently, insurers have been limited in their ability to monitor individuals' private investments and assess how these investments would impact losses in the event of a fire. Households lag well behind insurers in access to information about risk mitigation opportunities. Landscape features and appropriate risk mitigation tactics can vary widely from location to location, whereas public information about risk mitigation measures tends to be less specific.

Improvements in granular data collection (e.g., satellite imagery, drone observations) now

* Boomhower: Department of Economics, University of California San Diego, and NBER (email: jboomhower@ucsd.edu); Fowlie: Department of Agricultural and Resource Economics, University of California Berkeley, and NBER (email: fowlie@berkeley.edu); Plantinga: Bren School of Environmental Science and Management, University of California Santa Barbara (email: plantinga@ucsb.edu). We gratefully acknowledge funding from the UC Lab Fees program (grant ID LFR-20-652467) and the Carnegie Corporation of New York.

† Go to https://doi.org/10.1257/pandp.20231104 to visit the article page for additional materials and author disclosure statement(s).

make it possible to include structure-specific and community-level mitigation information in wildfire loss modeling. Recent analyses suggest that there are many opportunities for cost-effective risk mitigation in high-risk locations across the American West (Czajkowski et al. 2020). This suggestive evidence notwithstanding, some insurers have expressed skepticism that insurance premium discounts will be sufficient to increase private investment, citing high implementation costs and questionable efficacy. Firms have been reluctant to adjust premiums to reward risk mitigation efforts.[1] In California, regulators have a different view. In 2022, California became the first state to mandate that all insurers account for self-protection measures—and the reduction in risk therefrom—in their insurance premium pricing.[2]

Against this backdrop, we consider the channels through which risk-mitigation-based (RMB) discounts can increase private investments in self-protection. We extend the canonical theory to account for three salient features of the wildfire risk mitigation choice context: insurance mandates, information frictions, and postdisaster relief programs. We identify two channels through which RMB premium discounts can increase investments in self-protection. And we show how the government's optimal choice of disaster relief depends on the degree to which information about the risk-reducing effects of self-protection increases moral hazard.

I. Baseline Model

We consider a representative household's choice of wildfire risk mitigation r in a market characterized by an effective insurance mandate. In the United States, most mortgage lenders require homeowners to purchase homeowners insurance, which covers wildfires. Although a large majority of homeowners living in wildfire-prone areas have homeowners insurance, *under*insurance is common. Lenders do not typically require coverage for complete rebuilding costs, nor do they require coverage that accommodates inflation, "surge" pricing, or temporary housing. We thus assume that the level of insurance s that is required by lenders does not cover all costs of losing the home to wildfire.

Regulators routinely intervene in insurance markets to ensure the availability of insurance, the solvency of insurers, and fair insurance pricing. To assure solvency in the event of a catastrophic wildfire, insurers must build up reserves and/or purchase reinsurance because loss distributions are characterized by fat tails and highly correlated risks (Kousky 2019). This can increase the price of wildfire insurance above actuarially fair rates. We thus assume that regulated premia exceed actuarially fair rates by a factor of $\pi > 1$. At these prices, households insure up to, but not more than, the level required by their mortgage lender.

Insurers have historically lacked the ability to assess the relationships between individual protective measures and wildfire losses with any degree of precision. But their analytical capacity is evolving rapidly. We consider two regimes: one in which premia are independent of expenditures on self-protection and one in which insurers offer premium discounts that reflect the relationship between mitigation investments r and the risk of wildfire p.

For simplicity, we assume that an individual household is faced with two states of the world $(0, 1)$ with probabilities $1 - p$ and p. Real income endowments are denoted I_0^e and I_1^e, respectively. Let d denote the cost of a wildfire loss: $I_0^e - I_1^e = d$. Following Ehrlich and Becker (1972), households can invest a continuous amount r in self-protection (e.g., defensible space, fire-safe building upgrades) to reduce the risk of damage. We normalize the price of self-protection to 1 and assume that the probability of loss is $p(r), p'(r) < 0, p''(r) > 0$.

Disaster relief and other public transfers to households reduce the incentives for private investments in self-protection (Baylis and Boomhower 2023). In the model, government transfers to households in the loss state are represented by γ. All households pay a tax τ to

[1] Moorcraft, Bethan. 2022. "California's New Wildfire Safety Regulation 'a Big Burden for Insurers.'" *Insurance Business America*. December 12. https://www.insurancebusinessmag.com/us/news/catastrophe/californias-new-wildfire-safety-regulation-a-big-burden-for-insurers-430336.aspx.

[2] California's "Mitigation in Rating Plans and Wildfire Risk Models" (REG-2020-00015). http://www.insurance.ca.gov/0250-insurers/0500-legal-info/0300-workshop-insurers/upload/Mitigation-in-Rating-Plans-and-Wildfire-Risk-Models-Revised-Workshop-Draft-Text-of-Regulation.pdf.

cover the costs of these transfers. In Section III, we analyze how the government determines γ and τ.

We first consider the investment choice of a fully informed household. Under status quo insurance pricing, the binding insurance requirement imposes a fixed cost of πps. The household's problem is given by

$$\max_r \quad p(r)U\big(I_1^e - r + (1-\pi p)s + \gamma - \tau\big) + \big(1 - p(r)\big)U\big(I_0^e - r - \pi ps - \tau\big).$$

The first-order optimality condition for the choice of r^* is

$$(1) \quad \underbrace{-p'(r)\big(U_0 - U_1\big)}_{\text{marginal benefit}} = \underbrace{p\,U_1' + (1-p)\,U_0'}_{\text{marginal cost}},$$

where U_1' and U_0' denote marginal utility in the loss and no-loss states, respectively.

The marginal benefit of self-protection is the reduction in the probability of utility losses from wildfire damages, and the marginal cost is the probability-weighted average across states of marginal utility losses from investment expenditures. Because insurance and government transfers reduce the difference in utility across states, $U_0 - U_1$, marginal benefits are decreasing with s and γ. If damages are fully offset by insurance and relief payments, risk mitigation would confer no benefits to the household. However, we will show below that the government's optimal choice of γ leaves households less than fully insured.

II. Premium Discounts for Risk Mitigation

We now consider the implications of requiring insurers to offer premium discounts. Let $\delta\big(p(0) - p(r)\big)$ represent the RMB premium discount per dollar of coverage, where $\big(p(0) - p(r)\big)$ measures the risk reduction generated by private investments in risk mitigation r. An actuarially fair discount would set $\delta = 1$. Under the status quo (no discount), $\delta = 0$.

For ease of exposition, we let $X_1 \equiv I_1^e + s + \gamma - \pi s - \tau$ and $X_0 \equiv I_0^e - \pi s - \tau$. We use $D(r)$ to represent the discount schedule: $D(r) \equiv \delta\big(p(0) - p(r)\big)$. The household's problem is now given by

$$(2) \quad \max_r \quad p(r)U\big(X_1 - r + D(r)s\big) + \big(1 - p(r)\big)U\big(X_0 - r + D(r)s\big).$$

The first-order optimality condition for the choice of r^* is

$$(3) \quad -p'(r)\big(U_0 - U_1\big) = p\,U_1'\big(1 + \delta p'(r)s\big) + (1-p)\,U_0'\big(1 + \delta p'(r)s\big).$$

In contrast to equation (1), the private costs of risk mitigation are now partially offset by the premium discount.

Applying the implicit function theorem to this optimality condition, we obtain the marginal effect of the discount parameter on the optimal choice of self-protection:

$$(4) \quad \frac{dr}{d\delta} = \frac{-p'(r)\,U_0' - p^2U'' - p(1-p)\,U_0''}{-H} > 0,$$

where H denotes the Hessian matrix. The second-order condition for utility maximization ($H < 0$) implies that the denominator is positive. Under risk aversion, the second derivative of the utility function is negative. Thus, assuming an interior solution, investments in r increase with the premium discount as households internalize more of the returns on these risk mitigation investments.

Equation (2) ignores the cost implications of offering RMB discounts. Insurers will presumably recover these costs by raising the base premiums they charge. The extent to which this has an offsetting effect on r will depend on the scale of the costs and the concavity of the utility function.

A. *Information Frictions*

The preceding analysis has been predicated on the strong assumption that households are fully informed about their wildfire risk exposure and the available opportunities for self-protection. This assumption abstracts away from some first-order information frictions. Although advances in data quality and risk modeling are alleviating some of these frictions—for example, baseline wildfire risk scores are now readily

available to all[3]—it remains very challenging to discern how this baseline risk could be reduced by private investments in risk mitigation.

If a household wants to make an informed choice of r, they must incur the cost of collecting and processing complex information that pertains to their particular home. If these costs are high, they may rationally decide not to incur these costs. We consider a stylized model of a two-stage decision in which the household first chooses whether to remain with the default choice of $r = 0$ or acquire the information needed to make an informed choice. If the household incurs the costs of collecting and processing the available information i, the posterior beliefs about $p(r)$ informed by this information set are denoted $f(p(r); i)$.

The household's objective is to maximize the expectation of utility less the cost of collecting information. The household will choose to incur the cost of making an informed choice if they anticipate that

$$\begin{aligned} p(0)U(X_1) + \big(1 - p(0)\big)U(X_0) \\ < \max_r \int \Big[p(r)U\big(X_1 - r + D(r)s\big) \\ + \big(1 - p(r)\big)U\big(X_0 - r + D(r)s\big)\Big] \\ f(p(r); i)\, dp(r) - C(i). \end{aligned}$$

In this choice context, the introduction of RMB discounts can increase private investments in risk mitigation via two channels. First, as elucidated above, premium discounts provide a financial incentive, helping the household to internalize the benefits of investments that reduce insured damages. This incentive weakly increases the benefits of making an informed investment choice.

The introduction of RMB discounts can also increase r through an information channel. The premium schedule $D(r)$ summarizes valuable, property-specific information that has been difficult, if not impossible, for households to access previously. A household might reasonably use $D(r)$ to inform their posterior beliefs. In other words, the introduction of premium discounts could substantially lower the cost of accessing information $C(i)$.

Equipped with better information, underinsured households that have been underinvesting in r will have two reasons to increase their risk mitigation efforts. One is to take advantage of premium discounts (the price channel). The second is to pursue cost-effective opportunities to reduce uninsured risk (the information channel).

B. *Premium Discounts and Efficient Investment*

RMB premium discounts have the potential to increase investments in self-protection. The extent to which this potential is actually realized will depend in part on how discount schedules are presented to households. Insurance pricing is highly complex. Mapping changes in home attributes to changes in insurance premiums can require navigating many pages of actuarial tables and calculations. Echoing the observation made by Mulder and Kousky (2023) in the context of flood insurance, it will be important to insure that premium discount schedules are highly salient and easy to understand.

Even if these price and information channels are fully leveraged, RMB discounts will be insufficient to induce socially optimal levels of investment. One reason is that when there are positive spillovers to neighboring properties, the social benefits generated by private investments in risk mitigation are only partially captured by insurance companies and private households. Additional interventions will be needed to coordinate efficient adaptation investments at the community level.

III. Government Disaster Relief

Government relief payments in the loss state, γ, and the tax rate, τ, are exogenous to the household but are choice variables for the government. Adopting the household model with premium discounts in Section II, we investigate how the government would determine the optimal level of relief, assuming its objective is to maximize the aggregate utility of households. We assume, further, that the government accounts for the dependence of household self-protection, r, on its choices of γ and τ.

Following Chetty (2006), it is instructive to write the solution to the household's problem as

[3] See, for example, https://firststreet.org/risk-factor/fire-factor/.

$$V(\gamma,\tau) = \max_{\{I_0,I_1,\lambda_0,\lambda_1,r\}}(1-p)U(I_0) + pU(I_1) + \lambda_0\left(I_0^e + s\left(D(r)-\pi\right) - r - \tau - I_0\right) + \lambda_1\left(I_1^e + s\left(1+D(r)-\pi\right) - r + \gamma - \tau - I_1\right),$$

where I_0 and I_1 are net income in the no-loss and loss states and λ_0 and λ_1 are shadow prices on the income constraints. $V(\gamma,\tau)$ is maximized household utility conditional on disaster relief and the tax rate. Normalizing the number of households to 1, the government maximizes V subject to the balanced budget constraint $\tau = p(r)\gamma$. The budget constraint gives τ as a function of γ, and, thus the government's problem can be written

$$\max_\gamma V\left(\gamma,\tau(\gamma)\right). \tag{5}$$

Recognizing that V is already maximized with respect to $I_0, I_1, \lambda_0, \lambda_1$, and r, and applying the envelope theorem, the first-order condition for (5) is

$$\frac{dV}{d\gamma} = -\lambda_0\frac{d\tau}{d\gamma} + \lambda_1\left[1-\frac{d\tau}{d\gamma}\right] = 0. \tag{6}$$

From the government's budget constraint, $\frac{d\tau}{d\gamma} = p'(r)\frac{dr}{d\gamma}\gamma + p(r)$, and from the household's problem, $\lambda_0 = (1-p)U_0'$ and $\lambda_1 = pU_1'$. Substituting into (6) and rearranging yields

$$\frac{U_1'}{U_0'} = \frac{(1-p)\left[p'(r)\frac{dr}{d\gamma}\gamma + p(r)\right]}{p\left[1-p'(r)\frac{dr}{d\gamma}\gamma - p(r)\right]}. \tag{7}$$

According to (7), the government's optimal choice of γ depends on the sign of $\frac{dr}{d\gamma}$. If $\frac{dr}{d\gamma} < 0$, then $\frac{U_1'}{U_0'} > 1$, which by the concavity of the utility function implies $I_0 > I_1$. Comparing income in the loss and no-loss states, we find that $I_0 > I_1 \Leftrightarrow d > s + \gamma$. In words, when households reduce self-protection in response to an increase in disaster relief (i.e., there is moral hazard), the government's optimal level of relief will less than fully insure households against damages.

Alternatively, it is possible that $\frac{dr}{d\gamma} > 0$, in which case $I_1 > I_0$ and $s + \gamma > d$ by (7). This would imply that when households increase self-protection in response to disaster relief, the government makes them better off in the loss state. However, this solution can be ruled out by solving for $\frac{dr}{d\gamma}$ using the household's first-order condition in (3). In particular, it can be shown that if income is higher in the loss state ($I_1 > I_0$), the household will always decrease self-protection in response to more relief ($\frac{dr}{d\gamma} < 0$). In other words, the government cannot choose an optimal value of γ to make households better off in the loss state and induce them to increase self-protection.[4]

The final case is $\frac{dr}{d\gamma} = 0$. If the household does not change self-protection in response to the relief payment, then there is no moral hazard, and the government uses the transfer γ to fully insure households in the event of a disaster. One explanation for $\frac{dr}{d\gamma} = 0$ is that households have no information about the relationship $p(r)$. If the requirement for insurers to offer premium discounts provides this information, our model implies that households will reduce self-protection in response to disaster relief (i.e., $\frac{dr}{d\gamma} < 0$). In other words, RMB discounts can increase the moral hazard costs associated with disaster relief via the information channel.

IV. Conclusion

Insurance premiums that reward private investment in wildfire risk mitigation have a potentially important role to play in accelerating adaptation to escalating wildfire risk. We explore the implications of RMB discounts in a model that incorporates information frictions, incomplete insurance, and government relief payments to households impacted by wildfire.

Premium discounts can increase private investment through two channels. First, the discounts allow households to internalize the benefits of risk mitigation associated with reductions in insured losses. Second, premium discounts can generate additional value by communicating information about how investments in self-protection can reduce uninsured losses. To

[4] In contrast, for the first case mentioned above, there is an interior solution to the government's problem that is consistent with household behavior: $\frac{dr}{d\gamma} < 0$ when $I_0 > I_1$.

fully leverage either channel, discount schedules must be communicated in a way that households can easily understand.

Finally, we note that a reduction in information frictions can increase the moral hazard of publicly provided disaster relief payments. Taken together, our results highlight the importance of coordinating insurance market regulation, information provision, and government assistance.

REFERENCES

Baylis, Patrick W., and Judson Boomhower. 2021. "Mandated vs. Voluntary Adaptation to Natural Disasters: The Case of U.S. Wildfires." NBER Working Paper 29621.

Baylis, Patrick, and Judson Boomhower. 2023. "The Economic Incidence of Wildfire Suppression in the United States." *American Economic Journal: Applied Economics* 15 (1): 442–73.

Calkin, David E., Jack D. Cohen, Mark A. Finney, and Matthew P. Thompson. 2014. "How Risk Management Can Prevent Future Wildfire Disasters in the Wildland-Urban Interface." *Proceedings of the National Academy of Sciences* 111 (2): 746–51.

Chetty, Raj. 2006. "A General Formula for the Optimal Level of Social Insurance." *Journal of Public Economics* 90 (10–11): 1879–901.

Cohen, Jack D. 2000. "Preventing Disaster: Home Ignitability in the Wildland-Urban Interface." *Journal of Forestry* 98 (3): 15–21.

Czajkowski, Jeffrey, Michael Young, Ian Giammanco, Matt Nielsen, Elisabetta Russo, Anne Cope, Aaron Brandenburg, and Lisa Groshong. 2020. *Application of Wildfire Mitigation to Insured Property Exposure*. Kansas City, MO: Center for Insurance Policy Research, National Association of Insurance Commissioners.

Ehrlich, Isaac, and Gary S. Becker. 1972. "Market Insurance, Self-Insurance, and Self-Protection." *Journal of Political Economy* 80 (4): 623–48.

Goolsby, Julia B., Patricia A. Champ, Hannah Brenkert-Smith, Bobbi J. Clauson, Robert M. Sgroi, Lesley Williams, Christopher M. Barth, James R. Meldrum, Colleen Donovan, and Carolyn Wagner. 2022. *Living with Wildfire in Teton County, Wyoming: 2021 Data Report*. US Department of Agriculture Forest Service Rocky Mountain Research Station Research Note RMRS-RN-93.

Kousky, Carolyn. 2019. "The Role of Natural Disaster Insurance in Recovery and Risk Reduction." *Annual Review of Resource Economics* 11: 399–428.

Kunreuther, Howard C., and Erwann O. Michel-Kerjan. 2009. *At War with the Weather: Managing Large-Scale Risks in a New Era of Catastrophes.* Cambridge, MA: MIT Press.

Lueck, Dean, and Jonathan Yoder. 2016. "Clearing the Smoke from Wildfire Policy: An Economic Perspective." PERC Policy Series 56.

Mulder, Philip, and Carolyn Kousky. 2023. "Risk Rating without Risk Information." https://doi.org/10.1257/pandp.20231102.

Radeloff, Volker C., David P. Helmers, H. Anu Kramer, Miranda Mockrin, Patricia M. Alexandre, Avi Bar-Massada, Van Butsic, et al. 2018. "Rapid Growth of the US Wildland-Urban Interface Raises Wildfire Risk." *Proceedings of the National Academy of Sciences* 115 (13): 3314–19.

Shafran, Aric P. 2008. "Risk Externalities and the Problem of Wildfire Risk." *Journal of Urban Economics* 64 (2): 488–95.